Class 12 Mathematics

Chapter 6: Application of Derivatives

Practice Set A: step-wise solutions

20 questions **52** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 MCQ Board-pattern — 1 mark

The radius of a circular plate is increasing at the rate of 0.2 cm/s. The rate at which its area is increasing when the radius is 5 cm is:

(A) π cm^2/s **(B)** 2π cm^2/s **(C)** 10π cm^2/s **(D)** 25π cm^2/s

Formula / idea: Chain rule: $\dfrac{dA}{dt} = \dfrac{dA}{dr} \cdot \dfrac{dr}{dt}$

1 Area of a circle
$A = \pi r^2$

2 Differentiate both sides w.r.t. t
$\dfrac{dA}{dt} = 2\pi r \dfrac{dr}{dt}$

3 Put $r = 5$ and $\dfrac{dr}{dt} = 0.2$ ✓ 1
$\dfrac{dA}{dt} = 2\pi(5)(0.2) = 2\pi$ cm^2/s

Answer **(B)** 2π cm^2/s

Watch out: Forgetting to multiply by $\dfrac{dr}{dt}$ gives 10π.

2 MCQ CBSE SQP 2024-25 — 1 mark

The largest interval in which the function f defined by $f(x) = e^x$ is strictly increasing, is:

(A) $[1, \infty)$ **(B)** $(-\infty, 0)$ **(C)** $(-\infty, \infty)$ **(D)** $(0, \infty)$

Formula / idea: If $f'(x) > 0$ for all x in an interval, f is strictly increasing there.

1 Differentiate
$f'(x) = e^x$

2 $e^x > 0$ for every real x
$f'(x) > 0$ for all $x \in \mathbb{R}$

3 So f is strictly increasing on the whole real line
Largest interval $= (-\infty, \infty)$ ✓ 1

Answer (C) $(-\infty, \infty)$

Watch out: Picking (A), (B) or (D): f increases there too, but each is only a part of $\mathbb{R}$.

3 MCQ CBSE SQP 2025-26 1 mark

A function $f(x) = 10 - x - 2x^2$ is increasing on the interval:

(A) $\left(-\infty, -\frac{1}{4}\right]$ **(B)** $\left(-\infty, \frac{1}{4}\right)$ **(C)** $\left[-\frac{1}{4}, \infty\right)$ **(D)** $\left[-\frac{1}{4}, \frac{1}{4}\right]$

Formula / idea: f is increasing where $f'(x) \geq 0$.

1 Differentiate
$f'(x) = -1 - 4x$

2 Put $f'(x) \geq 0$
$-1 - 4x \geq 0 \Rightarrow 4x \leq -1$

3 Divide by 4
$x \leq -\frac{1}{4} \Rightarrow \left(-\infty, -\frac{1}{4}\right]$ ✓ 1

Answer (A) $\left(-\infty, -\frac{1}{4}\right]$

Watch out: Getting $x \geq -\frac{1}{4}$ by a sign slip, which gives (C), where f actually decreases.

4 MCQ CBSE 2025 1 mark

The absolute maximum value of the function $f(x) = x^3 - 3x + 2$ in $[0, 2]$ is:

(A) 0 **(B)** 2 **(C)** 4 **(D)** 5

Formula / idea: On $[a, b]$, compare f at the critical points and at the end points.

1 Differentiate and equate to zero
$f'(x) = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$

2 Only $x = 1$ lies in $[0, 2]$; find the values
$f(0) = 2, \quad f(1) = 0, \quad f(2) = 4$

3 Largest value

$\text{Absolute maximum} = 4 \text{ at } x = 2$ ✓ 1

Answer (C) 4

Watch out: Checking only the critical point $x = 1$ and forgetting the end points.

5 **MCQ** Board-pattern 1 mark

The local minimum value of $f(x) = x^2 + \dfrac{16}{x}$, $x > 0$, is:

(A) 2 **(B)** 8 **(C)** 12 **(D)** 16

Formula / idea: Second derivative test: $f'(c) = 0$ and $f''(c) > 0$ gives a local minimum at c.

1 Differentiate

$f'(x) = 2x - \dfrac{16}{x^2}$

2 Put $f'(x) = 0$

$2x^3 = 16 \Rightarrow x^3 = 8 \Rightarrow x = 2$

3 Second derivative test

$f''(x) = 2 + \dfrac{32}{x^3}, \quad f''(2) = 6 > 0$

4 So local minimum at $x = 2$; its value is ✓ 1

$f(2) = 4 + 8 = 12$

Answer (C) 12

Watch out: Giving the point $x = 2$ (option A) instead of the value $f(2) = 12$.

6 **Assertion-Reason** CBSE SQP 2023-24 1 mark

Let $f(x)$ be a polynomial function of degree 6 such that $\frac{d}{dx}\big(f(x)\big) = (x-1)^3(x-3)^2$. **Assertion (A):** $f(x)$ has a minimum at $x = 1$. **Reason (R):** When $\frac{d}{dx}\big(f(x)\big) < 0$ for all $x \in (a-h, a)$ and $\frac{d}{dx}\big(f(x)\big) > 0$ for all $x \in (a, a+h)$, where h is an infinitesimally small positive quantity, then $f(x)$ has a minimum at $x = a$, provided $f(x)$ is continuous at $x = a$.

(A) **Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: First derivative test: f' changes from $-$ to $+$ at c means local minimum at c.

1	Near $x = 1$, $(x-3)^2 > 0$ and $(x-1)^3$ changes sign from $-$ to $+$ $x < 1 : f'(x) < 0, \qquad x > 1 : f'(x) > 0$	
2	So f has a minimum at $x = 1$: **A is true**	
3	R is the first derivative test: **R is true**	
4	A is obtained by applying R at $a = 1$: R explains A Option (A)	✓ 1

Answer **(A) Both true; R is the correct explanation of A.**

Watch out: Thinking $(x-1)^3$ keeps one sign; an odd power changes sign at $x = 1$.

7 **Short answer** CBSE 2025 2 marks

The side of an equilateral triangle is increasing at the rate of 3 cm/s. At what rate is its area increasing when the side of the triangle is 15 cm?

Formula / idea: Area of an equilateral triangle of side x is $\frac{\sqrt{3}}{4}x^2$.

1	Let side $= x$; write the area $A = \frac{\sqrt{3}}{4}x^2$	✓ ½
2	Differentiate w.r.t. t $\frac{dA}{dt} = \frac{\sqrt{3}}{4} \cdot 2x \frac{dx}{dt}$	✓ ½

3 Put $x = 15$, $\dfrac{dx}{dt} = 3$

$$\frac{dA}{dt} = \frac{\sqrt{3}}{4} \cdot 2(15)(3) = \frac{45\sqrt{3}}{2}$$

✓ ½

4 Hence the area increases at

$$\frac{45\sqrt{3}}{2}\ \text{cm}^2/\text{s}$$

✓ ½

Answer **Area increases at** $\dfrac{45\sqrt{3}}{2}\ \text{cm}^2/\text{s}$

Watch out: Dropping the $\dfrac{dx}{dt} = 3$ factor, or writing the units as cm/s.

Level 2

19 marks

8 **Short answer** CBSE 2025 3 marks

Find the intervals in which the function $f(x) = 5x^3 - 3x^2$ is (i) increasing (ii) decreasing.

Formula / idea: $f'(x) \geq 0$ gives increasing; $f'(x) \leq 0$ gives decreasing.

1 Differentiate and factorise

$$f'(x) = 15x^2 - 6x = 3x(5x - 2)$$

✓ 1

2 Put $f'(x) = 0$

$$x = 0, \quad x = \frac{2}{5}$$

✓ ½

3 Check the sign of $f'(x)$ in each interval

$$x < 0 : f'(x) > 0$$
$$0 < x < \tfrac{2}{5} : f'(x) < 0$$
$$x > \tfrac{2}{5} : f'(x) > 0$$

✓ ½

4 (i) f is increasing in

$$(-\infty, 0] \cup \left[\frac{2}{5}, \infty\right)$$

✓ ½

5 (ii) f is decreasing in

$$\left[0, \frac{2}{5}\right]$$

✓ ½

Answer **(i) Increasing in** $(-\infty, 0] \cup \left[\dfrac{2}{5}, \infty\right)$ **(ii) Decreasing in** $\left[0, \dfrac{2}{5}\right]$

Watch out: Skipping the sign check: $f'(x) < 0$ between 0 and $\dfrac{2}{5}$, so f falls there.

9 **Short answer** CBSE SQP 2025-26 2 marks

A spherical ball of ice melts in such a way that the rate at which its volume decreases at any instant is directly proportional to its surface area. Prove that the radius of the ice ball decreases at a constant rate.

Formula / idea: $V = \frac{4}{3}\pi r^3$, $S = 4\pi r^2$, and $\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$

1	Let radius $= r$ at time t $V = \frac{4}{3}\pi r^3, \qquad S = 4\pi r^2$	
2	Given: volume decreases at a rate proportional to S $(k > 0)$ $\frac{dV}{dt} = -kS = -4k\pi r^2$	✓ ½
3	Differentiate V w.r.t. t $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$	✓ ½
4	Equate the two expressions $4\pi r^2 \frac{dr}{dt} = -4k\pi r^2$	✓ ½
5	Cancel $4\pi r^2$ on both sides $\frac{dr}{dt} = -k$ (constant)	✓ ½
6	Hence the radius decreases at the constant rate k	

Answer $\frac{dr}{dt} = -k$**, a constant; so the radius decreases at a constant rate.**

Watch out: Leaving out the minus sign: the volume is decreasing, so $\frac{dV}{dt}$ is negative.

10 **Short answer** PYQ-style 3 marks

Find the local maximum and local minimum values of $f(x) = 3x^4 + 4x^3 - 12x^2 + 12$.

Formula / idea: Second derivative test: $f''(c) < 0$ gives local max, $f''(c) > 0$ gives local min.

1	Differentiate and factorise $f'(x) = 12x^3 + 12x^2 - 24x = 12x(x+2)(x-1)$	✓ ½
2	Put $f'(x) = 0$ $x = -2,\ 0,\ 1$	✓ ½
3	Second derivative $f''(x) = 36x^2 + 24x - 24$	✓ ½
4	At $x = -2$: $f''(-2) = 72 > 0$, local minimum $f(-2) = 48 - 32 - 48 + 12 = -20$	✓ ½

5	At $x = 0$: $f''(0) = -24 < 0$, local maximum $f(0) = 12$	✓ ½
6	At $x = 1$: $f''(1) = 36 > 0$, local minimum $f(1) = 3 + 4 - 12 + 12 = 7$	✓ ½
7	Hence local maximum value $= 12$; local minimum values $= -20$ and 7	

Answer **Local maximum value 12 (at $x = 0$); local minimum values -20 (at $x = -2$) and 7 (at $x = 1$)**

Watch out: Missing the root $x = 0$ after taking $12x$ common.

11 **Short answer** CBSE SQP 2023-24 3 marks

Find the interval/s in which the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = xe^x$ is increasing.

Formula / idea: Product rule: $(uv)' = u'v + uv'$; f increases where $f'(x) \geq 0$.

1	Differentiate by the product rule $f'(x) = e^x + xe^x = e^x(x + 1)$	✓ 1
2	Put $f'(x) = 0$; e^x is never zero $x + 1 = 0 \Rightarrow x = -1$	✓ ½
3	Sign of $f'(x)$, using $e^x > 0$ $x > -1 : f'(x) > 0$ $x < -1 : f'(x) < 0$	✓ 1
4	Hence f is increasing in $[-1, \infty)$	✓ ½

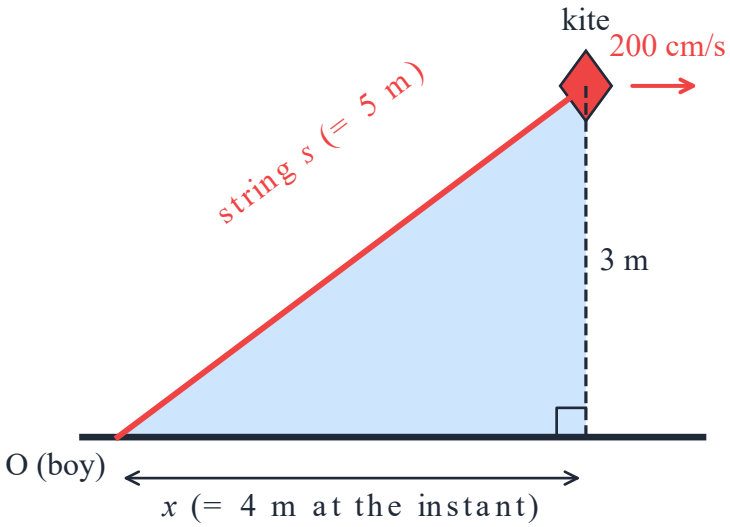
Answer **f is increasing in $[-1, \infty)$**

Watch out: Trying to solve $e^x = 0$; it has no solution, so only $x + 1$ decides the sign.

12 **Short answer** CBSE SQP 2024-25 3 marks

A kite is flying at a height of 3 metres and 5 metres of string is out. If the kite is moving away horizontally at the rate of 200 cm/s, find the rate at which the string is being released.

Formula / idea: Pythagoras theorem, then differentiate w.r.t. t.



1	Let horizontal distance $= x$ m and string $= s$ m; height is fixed at 3 m $s^2 = x^2 + 3^2$	✓ ½
2	When $s = 5$ $x = \sqrt{25 - 9} = 4$ m	✓ ½
3	Differentiate w.r.t. t $2s\dfrac{ds}{dt} = 2x\dfrac{dx}{dt}$	✓ 1
4	Put $x = 4, s = 5, \dfrac{dx}{dt} = 200$ cm/s $= 2$ m/s $\dfrac{ds}{dt} = \dfrac{x}{s}\cdot\dfrac{dx}{dt} = \dfrac{4\times 2}{5} = 1.6$ m/s	✓ ½
5	Hence the string is released at 1.6 m/s, i.e. 160 cm/s	✓ ½

Answer **String is released at 1.6 m/s ($= 160$ cm/s)**

Watch out: Mixing cm/s with metres; convert 200 cm/s to 2 m/s first.

13 **Short answer** CBSE 2025 3 marks

Find the absolute maximum and absolute minimum of the function $f(x) = 2x^3 - 15x^2 + 36x + 1$ on $[1, 5]$.

Formula / idea: On $[a, b]$, compare f at all critical points and at both end points.

1	Differentiate and factorise $f'(x) = 6x^2 - 30x + 36 = 6(x - 2)(x - 3)$	✓ ½
2	Critical points (both lie in $[1, 5]$) $x = 2, \quad x = 3$	✓ ½
3	Values at critical points and end points $f(1) = 24, \quad f(2) = 29$ $f(3) = 28, \quad f(5) = 56$	✓ 1
4	Largest value $\text{Absolute maximum} = 56 \text{ at } x = 5$	✓ ½
5	Smallest value $\text{Absolute minimum} = 24 \text{ at } x = 1$	✓ ½

Answer **Absolute maximum** $= 56$ **at** $x = 5$**; absolute minimum** $= 24$ **at** $x = 1$

Watch out: Taking the local max $f(2) = 29$ as the absolute maximum without checking $f(5)$.

14 **Short answer** CBSE SQP 2023-24 2 marks

Find the maximum profit that a company can make, if the profit function is given by $P(x) = 72 + 42x - x^2$, where x is the number of units and P is the profit in rupees.

Formula / idea: $P'(x) = 0$ and $P''(x) < 0$ give the maximum.

1	Differentiate $P'(x) = 42 - 2x$	✓ ½
2	Put $P'(x) = 0$ $x = 21$	✓ ½
3	Second derivative test $P''(x) = -2 < 0 \Rightarrow \text{maximum at } x = 21$	✓ ½

4 Maximum profit

$P(21) = 72 + 882 - 441 = ₹\,513$

✓ ½

Answer **Maximum profit** $= ₹\,513$ **(at** $x = 21$ **units)**

Watch out: Stopping at $x = 21$; the question asks for the profit $P(21)$.

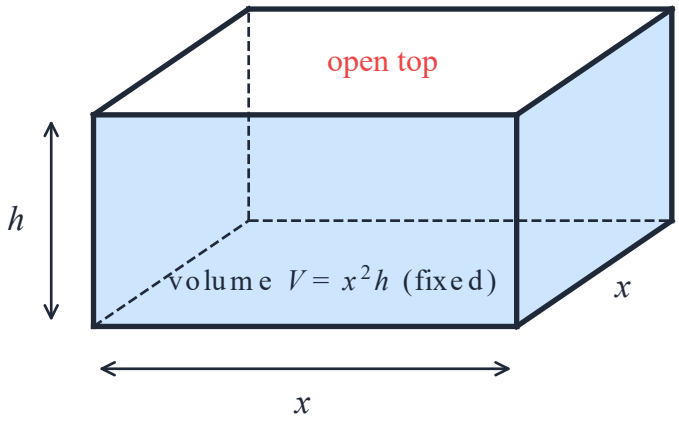
Level 3

25 marks

15 **Long answer** PYQ-style 5 marks

An open tank with a square base and vertical sides is to be constructed from a metal sheet so as to hold a given quantity of water. Show that the cost of the material will be least when the depth of the tank is half of its width.

Formula / idea: Cost of material $\propto$ area of sheet used; write the area in one variable using the fixed volume $V = x^2h$.



1 Let side of base $= x$, depth $= h$, fixed volume $= V$

$V = x^2h \Rightarrow h = \dfrac{V}{x^2}$

✓ 1

2 Area of sheet = base + four sides (no top)

$S = x^2 + 4xh$

Cost is proportional to this area

✓ ½

3 Put $h = \dfrac{V}{x^2}$

$S = x^2 + \dfrac{4V}{x}$

✓ ½

4 Differentiate w.r.t. x

$\dfrac{dS}{dx} = 2x - \dfrac{4V}{x^2}$

✓ ½

5	Put $\dfrac{dS}{dx} = 0$ $2x^3 = 4V \;\Rightarrow\; V = \dfrac{x^3}{2}$	✓ ½
6	Second derivative test, using $V = \dfrac{x^3}{2}$ $\dfrac{d^2S}{dx^2} = 2 + \dfrac{8V}{x^3} = 2 + 4 = 6 > 0$	✓ 1
7	So S is least here; find the depth $h = \dfrac{V}{x^2} = \dfrac{x^3/2}{x^2} = \dfrac{x}{2}$	✓ ½
8	Hence the cost is least when depth = half the width $h = \dfrac{x}{2}$	✓ ½

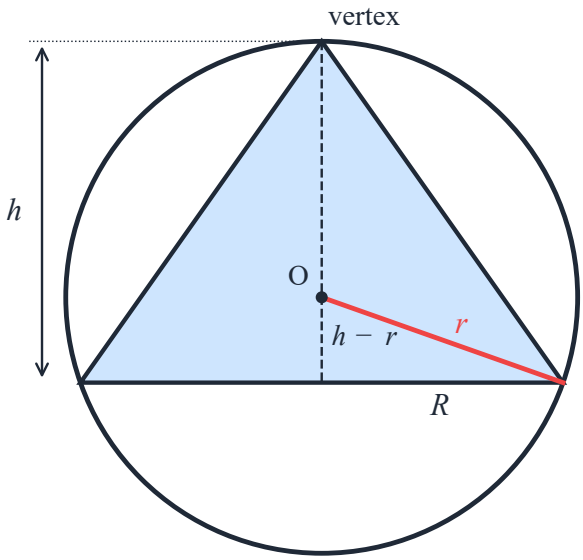
Answer **Sheet area (hence cost) is least when $h = \dfrac{x}{2}$, i.e. depth = half the width**

Watch out: Counting a top face in the area: the tank is open, so the sheet is $x^2 + 4xh$, not $2x^2 + 4xh$.

16 **Long answer** PYQ-style 5 marks

Show that the altitude of the right circular cone of maximum volume that can be inscribed in a sphere of radius r is $\dfrac{4r}{3}$. Also show that the maximum volume of the cone is $\dfrac{8}{27}$ of the volume of the sphere.

Formula / idea: Write V in one variable h, then use $V'(h) = 0$ and $V''(h) < 0$.



sphere of radius r (cross-section)

1	Cone: height h, base radius R; right triangle with hypotenuse r gives $R^2 + (h - r)^2 = r^2 \;\Rightarrow\; R^2 = 2rh - h^2$	✓ 1

2 Volume of the cone ✓ ½

$$V = \frac{1}{3}\pi R^2 h = \frac{\pi}{3}(2rh^2 - h^3)$$

3 Differentiate w.r.t. h ✓ ½

$$\frac{dV}{dh} = \frac{\pi}{3}(4rh - 3h^2)$$

4 Put $\frac{dV}{dh} = 0$ $(h \neq 0)$ ✓ ½

$$4r = 3h \;\Rightarrow\; h = \frac{4r}{3}$$

5 Second derivative at $h = \frac{4r}{3}$ ✓ 1

$$\frac{d^2V}{dh^2} = \frac{\pi}{3}(4r - 6h) = \frac{\pi}{3}(4r - 8r) = -\frac{4\pi r}{3} < 0$$

6 So V is maximum at altitude $h = \frac{4r}{3}$; then ✓ ½

$$R^2 = 2r \cdot \frac{4r}{3} - \frac{16r^2}{9} = \frac{8r^2}{9}$$

7 Maximum volume ✓ ½

$$V = \frac{1}{3}\pi \cdot \frac{8r^2}{9} \cdot \frac{4r}{3} = \frac{32\pi r^3}{81}$$

8 Compare with the sphere's volume $\frac{4}{3}\pi r^3$ ✓ ½

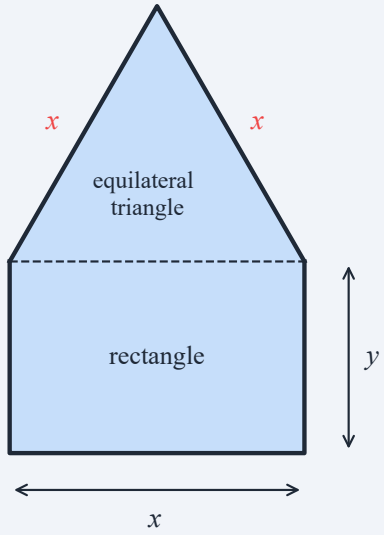
$$\frac{32\pi r^3/81}{4\pi r^3/3} = \frac{32}{81} \cdot \frac{3}{4} = \frac{8}{27}$$

Answer **Altitude** $= \frac{4r}{3}$**; maximum volume** $= \frac{32\pi r^3}{81} = \frac{8}{27} \times \frac{4}{3}\pi r^3$

Watch out: Taking the distance from the centre to the base as h instead of $h - r$.

17 **Long answer** PYQ-style 5 marks

A window is in the form of a rectangle surmounted by an equilateral triangle. The perimeter of the window is 12 m. Find the dimensions of the rectangle that will produce the largest area of the window.



Formula / idea: Use the perimeter to write area in one variable, then $A'(x) = 0$, $A''(x) < 0$.

1	Let width $= x$, height $= y$. Perimeter = base + 2 heights + 2 triangle sides $x + 2y + 2x = 12 \Rightarrow y = \dfrac{12 - 3x}{2}$	✓ 1
2	Area of window = rectangle + equilateral triangle $A = xy + \dfrac{\sqrt{3}}{4}x^2$	✓ ½
3	Substitute y $A = 6x - \dfrac{3}{2}x^2 + \dfrac{\sqrt{3}}{4}x^2$	✓ ½
4	Differentiate w.r.t. x $\dfrac{dA}{dx} = 6 - 3x + \dfrac{\sqrt{3}}{2}x$	✓ ½
5	Put $\dfrac{dA}{dx} = 0$ $x\left(3 - \dfrac{\sqrt{3}}{2}\right) = 6 \Rightarrow x = \dfrac{12}{6 - \sqrt{3}}$	✓ ½
6	Second derivative test $\dfrac{d^2A}{dx^2} = -3 + \dfrac{\sqrt{3}}{2} < 0 \Rightarrow \text{maximum}$	✓ 1
7	Find y $y = 6 - \dfrac{3}{2} \cdot \dfrac{12}{6 - \sqrt{3}} = \dfrac{18 - 6\sqrt{3}}{6 - \sqrt{3}}$	✓ ½

8 Hence, for the largest area, the rectangle has

$$\text{width} = \frac{4(6+\sqrt{3})}{11}\ \text{m}, \quad \text{height} = \frac{30-6\sqrt{3}}{11}\ \text{m}$$

Rationalise: multiply by $\frac{6+\sqrt{3}}{6+\sqrt{3}}$ ✓ ½

Answer **Width** $= \frac{12}{6-\sqrt{3}} = \frac{4(6+\sqrt{3})}{11}$ **m** (≈ 2.81 **m**), **height** $= \frac{18-6\sqrt{3}}{6-\sqrt{3}} = \frac{30-6\sqrt{3}}{11}$ **m** (≈ 1.78 **m**)

Watch out: Counting the common side (top of the rectangle) in the perimeter: it is not on the boundary.

18 **Short answer** PYQ-style 3 marks

Find the intervals in which the function $f(x) = \sin^4 x + \cos^4 x$, $0 \le x \le \frac{\pi}{2}$, is strictly increasing or strictly decreasing.

Formula / idea: $2\sin x\cos x = \sin 2x$, $\cos^2 x - \sin^2 x = \cos 2x$, $2\sin 2x\cos 2x = \sin 4x$

1 Differentiate

$f'(x) = 4\sin^3 x\cos x - 4\cos^3 x\sin x$ ✓ ½

2 Take $-4\sin x\cos x$ common and use double-angle formulas

$f'(x) = -4\sin x\cos x(\cos^2 x - \sin^2 x) = -2\sin 2x\cos 2x = -\sin 4x$ ✓ ½

3 Put $f'(x) = 0$ for $0 < x < \frac{\pi}{2}$, i.e. $0 < 4x < 2\pi$

$\sin 4x = 0 \Rightarrow 4x = \pi \Rightarrow x = \frac{\pi}{4}$ ✓ ½

4 For $0 < x < \frac{\pi}{4}$: $0 < 4x < \pi$, so $\sin 4x > 0$

$f'(x) < 0 \Rightarrow$ strictly decreasing in $\left(0, \frac{\pi}{4}\right)$ ✓ 1

5 For $\frac{\pi}{4} < x < \frac{\pi}{2}$: $\pi < 4x < 2\pi$, so $\sin 4x < 0$

$f'(x) > 0 \Rightarrow$ strictly increasing in $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ ✓ ½

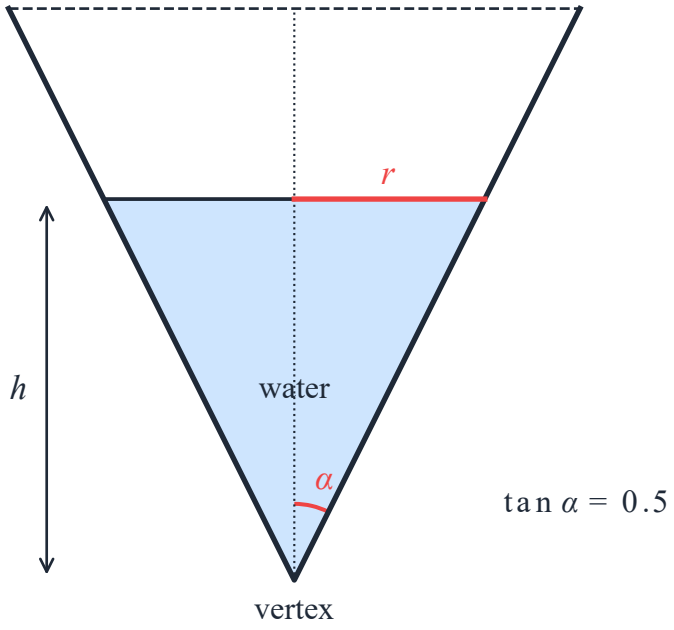
Answer **Strictly decreasing in** $\left(0, \frac{\pi}{4}\right)$**; strictly increasing in** $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

Watch out: Forgetting the minus sign in $f'(x) = -\sin 4x$, which swaps the two answers.

19 **Short answer** PYQ-style 3 marks

A water tank has the shape of an inverted right circular cone with its axis vertical and vertex lowermost. Its semi-vertical angle is $\tan^{-1}(0.5)$. Water is poured into it at a constant rate of $5\ \text{m}^3/\text{h}$. Find the rate at which the level of the water is rising at the instant when the depth of water in the tank is 4 m.

Formula / idea: Similar triangles fix r in terms of h; then $V = \dfrac{1}{3}\pi r^2 h$ in one variable.



1	Let depth $= h$ and radius of water surface $= r$; semi-vertical angle α has $\tan\alpha = 0.5$ $\dfrac{r}{h} = 0.5 \Rightarrow r = \dfrac{h}{2}$	✓ ½
2	Volume of water in terms of h $V = \dfrac{1}{3}\pi\left(\dfrac{h}{2}\right)^2 h = \dfrac{\pi h^3}{12}$	✓ ½
3	Differentiate w.r.t. t $\dfrac{dV}{dt} = \dfrac{\pi h^2}{4}\dfrac{dh}{dt}$	✓ 1
4	Put $\dfrac{dV}{dt} = 5$, $h = 4$ $5 = \dfrac{\pi(16)}{4}\dfrac{dh}{dt} \Rightarrow \dfrac{dh}{dt} = \dfrac{5}{4\pi}$	✓ ½
5	Hence the water level rises at $\dfrac{5}{4\pi}\ \text{m/h}$	✓ ½

Answer **The water level rises at $\dfrac{5}{4\pi}$ m/h (≈ 0.40 m/h)**

Watch out: Differentiating $V = \dfrac{1}{3}\pi r^2 h$ with both r and h varying; replace r by $\dfrac{h}{2}$ first.

20 **Case study** CBSE SQP 2025-26 4 marks

Case Study: LED bulbs are energy-efficient because they use much less electricity than traditional bulbs while producing the same amount of light. A company manufactures a new type of energy-efficient LED bulb. The cost of production and the revenue generated by selling x bulbs (in an hour) are modelled as $C(x) = 0.5x^2 - 10x + 150$ and $R(x) = -0.3x^2 + 20x$ respectively, where $C(x)$ and $R(x)$ are both in ₹. Use the given models to answer the following questions.

(i) Derive the profit function $P(x)$. 1 mark

(ii) Find the critical points of $P(x)$. 1 mark

(iii)(a) Determine whether the critical points correspond to a maximum or a minimum profit by using the second derivative test. 2 marks

OR

(iii)(b) Identify the possible practical value of x (i.e. the number of bulbs that can realistically be produced and sold) that can maximise the profit, if the resources available and the expenditure on machines allow the company to produce a minimum of 10 but not more than 18 bulbs per hour. Also calculate the maximum profit. 2 marks

Formula / idea: Profit = Revenue − Cost; $P'(x) = 0$ gives critical points; $P''(x) < 0$ gives a maximum.

(i) 1 mark

1 Profit = Revenue − Cost
$P(x) = R(x) - C(x) = (-0.3x^2 + 20x) - (0.5x^2 - 10x + 150)$ ✓ ½

2 Simplify
$P(x) = -0.8x^2 + 30x - 150$ ✓ ½

Answer (i) $P(x) = -0.8x^2 + 30x - 150$

(ii) 1 mark

1 Differentiate
$P'(x) = -1.6x + 30$ ✓ ½

2 Put $P'(x) = 0$
$x = \dfrac{30}{1.6} = 18.75$ ✓ ½

Answer (ii) **Only critical point:** $x = 18.75$

(iii)(a) 2 marks

1 Second derivative
$P''(x) = -1.6$ ✓ 1

2 At $x = 18.75$, $P'' < 0$. Hence the critical point gives maximum profit
$P''(18.75) = -1.6 < 0 \Rightarrow \text{maximum}$ ✓ 1

Answer (iii)(a) $P''(18.75) = -1.6 < 0$, **so** $x = 18.75$ **gives maximum profit**

OR

(iii)(b) 2 marks

1	For $x < 18.75$, $P'(x) > 0$, so P is increasing on $[10, 18]$ $P'(x) = -1.6x + 30 > 0$ for $10 \leq x \leq 18$	✓ 1
2	So profit is greatest at the largest allowed value $x = 18$ bulbs per hour	✓ ½
3	Maximum profit $P(18) = -0.8(324) + 540 - 150 = 130.8$	✓ ½

Answer (iii)(b) $x = 18$ **bulbs per hour; maximum profit** $= ₹\,130.80$

Watch out: In (iii)(b), choosing $x = 19$ (nearest whole number to 18.75); only $10 \leq x \leq 18$ is allowed.